

# Introduction to CSS

Learn how CSS styles, lays out, and enhances HTML pages — from basic styling to the cascade.

## ■ What is CSS?

CSS (Cascading Style Sheets) is used to style and lay out web pages, including colors, fonts, spacing, and positioning of elements. While HTML gives structure to a web page, CSS makes it look beautiful and usable.

## Why CSS?

Without CSS, websites would look plain, like unstyled documents. CSS helps you:

- Change colors and fonts
- Add spacing and layout
- Make responsive designs for mobile
- Animate and transition between states
- Separate concerns: HTML for structure, CSS for style

## ■ How CSS Works with HTML

CSS can be applied to HTML in three main ways:

### 1. Inline CSS

CSS written inside an HTML tag using the **style** attribute.

```
<p style="color: blue; font-size: 18px;">
  This is a blue paragraph.
</p>
```

### 2. Internal CSS

CSS written inside a **<style>** tag within the **<head>** section of the HTML.

```
<style>
p {
  color: green;
  font-weight: bold;
}
</style>
```

### 3. External CSS (Best Practice)

CSS written in a separate file and linked to the HTML file. The source describes this as the most recommended method for real-world projects.

#### Typical files

| File       | Purpose                      |
|------------|------------------------------|
| style.css  | Contains CSS rules.          |
| index.html | Contains the HTML structure. |

#### style.css

```
h1 {  
  color: darkred;  
  text-align: center;  
}
```

#### index.html

```
<!DOCTYPE html>  
<html>  
<head>  
  <link rel="stylesheet" href="style.css">  
</head>  
<body>  
  <h1>Welcome to CSS</h1>  
</body>  
</html>
```

### ■ The “Cascading” in CSS

If there are multiple rules targeting the same element, CSS uses the cascade to decide which rule to apply. The source identifies these factors:

| Factor              | Meaning                       |
|---------------------|-------------------------------|
| Specificity         | How specific the selector is. |
| Order of appearance | Which rule appears later.     |
| Importance          | The use of <b>!important</b>. |

#### Example

```
<p style="color: red;">  
  This will be red because inline CSS wins.  
</p>
```

## ■ Anatomy of a CSS Rule

A CSS rule is made of a selector, one or more properties, and their values.

```
selector {  
  property: value;  
}  
  
p {  
  color: black;  
  font-size: 16px;  
}
```

| Part     | Example          | Role                                         |
|----------|------------------|----------------------------------------------|
| Selector | p                | Targets all <code>&lt;p&gt;</code> elements. |
| Property | color, font-size | Defines what aspect is being styled.         |
| Value    | black, 16px      | Specifies the setting for each property.     |

## ■ What You'll Learn in CSS

As you move forward, you'll learn how to:

- Style text, backgrounds, and borders
- Control layout with Flexbox and Grid
- Make your website responsive and mobile-friendly
- Animate elements and transitions
- Write modern, maintainable CSS

### ■ Tip

- Keep HTML focused on structure and CSS focused on presentation.
- For real-world projects, the source recommends separating CSS into an external stylesheet.
- Understanding the cascade is essential when multiple CSS rules target the same element.

## Quick Takeaway

**HTML** provides structure; **CSS** controls presentation. CSS can be written inline, internally, or externally. A CSS rule consists of a **selector**, **property**, and **value**, while the cascade determines which competing rule is applied.